

ITEC622 – Data Analytics and Visualization

Assessment 3: Data Analytics and Visualization Case Study

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Part A: Data Analytics and Visualization Proposal for ABC Environment

1. Introduction

ABC Environment is a worldwide non-profit organization focused on environmental protection and sustainable development. With mounting climate change impacts and environmental degradation, the organization has grown quickly over the past couple of years using a diverse network of volunteers throughout the continent. To maintain its benefits, make data-driven choices, and track its activity, ABC Environment needs a comprehensive data analytics and visualization system. This proposal will describe the development and implementation of that system to support goals, provide operational transparency, and encourage environmental and social well-being.

2. Project Objectives

The primary objective of this project is to develop a data analytics and visualization solution that empowers ABC Environment to make informed, ethical, and strategic decisions. Specifically, the project aims to:

- Track and lower the environmental footprint of the organization.
- Assess and improve engagement with volunteers in a variety of locations and demographics.
- Measure donor contributions and fundraising outcomes to increase financial sustainability.
- Assess outcomes of project work and the effect of environmental initiatives made globally.
- Provide transparency and accountability with public dashboards and reports.

3. Data Sources and Data Collection Tools

3.1. Data Requirements

The solution will require both internal and external datasets to cover the following domains:

- Environmental metrics
- Volunteer records

3.2. Data Sources

Source Type	Examples	Purpose
Internal	Volunteer and donor databases, project reports	Operational performance, fundraising, engagement
External	UN Climate Change Data, Kaggle datasets, World Bank indicators	Environmental context, benchmarking

Table 1: Data sources of CO2 emission data.

3.3. Data Collection Tools and Techniques

- **Internal Data Collection:**
 - Online forms (Google Forms, Microsoft Forms)
 - CRM systems (e.g., Salesforce for NGOs)
 - Automated logging from internal apps or volunteer portals
- **External Data Collection:**
 - Web scraping
 - APIs (World Bank)
 - Downloads from open data portals (Kaggle)

3.4. Data Acquisition and Retention

- We will obtain all data using batch imports and scheduled API requests.
- All data will be kept securely in a cloud-based environment.
- All retention policies will follow GDPR and ethical data use norms and protocols for anonymization of personal data will be in place.

4. Analytics and Visualization Tools/Techniques

4.1. Cleaning and Transformation Tools

- **Microsoft Excel** – For preliminary cleaning, validation, importing, and transforming.
- **Power Query (Power BI)** – For shaping data and visualizing it.

4.2. Analysis Techniques

- Descriptive statistics
- Time-series analysis
- Geographic analysis

4.3. Visualization Tools and Best Practices

- **Power BI**: Chosen for its ability to create interactive dashboards.
- **Chart Types**: Stacked bar chart, ribbon charts, pie charts, and maps.
- Visualizations will follow the 4Cs principles:
 - **Clear**: Simplified axes and titles.
 - **Clean**: Minimal visual clutter.

- Concise: Only essential metrics.
- Captivating: Use of color and interactivity to draw attention.

The solution will be scalable and user-friendly, allowing decision-makers and the public to interact with the data through an intuitive interface.

5. Use of Data Visualization for Preservation of the Common Good

This project provides a direct benefit to the common good through enhancing ABC Environment's capacity to:

- Facilitate environmentally sustainable change through evidence-based decision making,
- Engage communities and volunteers by offering performance information that is inclusive and collaborative,
- Increase transparency and demonstrate to donors their contributions.
- Learn about disadvantaged areas or causes that would be better served by initiatives with the greatest social and environmental benefit.

By following ethical data practices, the proposed system will act as a model for other social NGO organizations using analytics to further their mission.

Conclusion

As we focus on making data-driven decisions every day, ABC Environment has an unparalleled opportunity to use analytics and visualization to increase its impact. This proposal has provided a robust solution that exemplifies its commitment to preserving the environment, empowering communities, and providing transparency. After absorbing the suggestions provided in the relevant tools, datasets, and techniques, ABC Environment will be able to generate better strategic outcomes, build confidence with stakeholders, and promote sustainable practices throughout the world.

Part B: Pilot Visualization

Dataset 1: Global CO2 Emissions

Data Source	World Development Indicators	1960	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974	1975
Aruba	ABW	Carbon di	EN.GHG.CO2.PC.CE.AR5									0.427481	0.489954	0.680483	0.759876	0.745514	0.984647
Africa Eas	AFE	Carbon di	EN.GHG.CO2.PC.CE.AR5									1.355303	1.308637	1.321972	1.401468	1.397943	1.421605
Afghanistan	AFG	Carbon di	EN.GHG.CO2.PC.CE.AR5									0.153577	0.149875	0.142875	0.142614	0.175658	0.158839
Africa Wes	AFW	Carbon di	EN.GHG.CO2.PC.CE.AR5									0.347427	0.457251	0.550165	0.697791	0.843933	0.658421
Angola	AGO	Carbon di	EN.GHG.CO2.PC.CE.AR5									1.526435	1.422026	1.678921	1.776153	1.785263	1.593568
Albania	ALB	Carbon di	EN.GHG.CO2.PC.CE.AR5									2.268812	2.211346	2.460094	2.156045	2.267795	2.255502
Andorra	AND	Carbon di	EN.GHG.CO2.PC.CE.AR5														
Arab World	ARB	Carbon di	EN.GHG.CO2.PC.CE.AR5									2.247111	2.332057	2.53297	2.84615	2.704652	2.63361
United Arab Emirates	ARE	Carbon di	EN.GHG.CO2.PC.CE.AR5									67.93143	75.90373	72.06855	82.13493	73.62841	67.56746
Argentina	ARG	Carbon di	EN.GHG.CO2.PC.CE.AR5									3.667305	3.623286	3.640408	3.705198	3.6636	3.51381
Armenia	ARM	Carbon di	EN.GHG.CO2.PC.CE.AR5									4.079563	4.005591	4.202005	4.473252	4.720811	5.045159
American Samoa	ASM	Carbon di	EN.GHG.CO2.PC.CE.AR5									0.0037	0.003632	0.003572	0.003517	0.003463	0.003403
Antigua and Barbuda	ATG	Carbon di	EN.GHG.CO2.PC.CE.AR5									1.586908	1.580435	1.928345	1.953623	1.746854	2.122527
Australia	AUS	Carbon di	EN.GHG.CO2.PC.CE.AR5									12.82297	12.53055	12.74622	13.2525	13.61556	13.93472
Austria	AUT	Carbon di	EN.GHG.CO2.PC.CE.AR5									7.093423	7.417297	7.667624	8.06157	7.720398	7.43415
Azerbaijan	AZE	Carbon di	EN.GHG.CO2.PC.CE.AR5									4.684158	4.590144	4.798524	5.105062	5.403693	5.750538

Figure 1: CO2 Emission dataset (World Bank Group, 2023)

a. Who collected the data?

- Collected by the **World Bank**, based on UNFCCC (United Nations Framework Convention on Climate Change) data.
- Link: [Carbon dioxide \(CO2\) emissions excluding LULUCF per capita \(t CO2e/capita\) | Data](#)

b. Variables included:

- Country Name
- Country Code

- Year
- CO₂ emissions (metric tons per capita)

c. Size of dataset:

- 9500 records
- Variables: 4 columns

d. Locations included:

- Global

e. Relevance and Limitations:

- **Supports:** Tracking environmental footprint per nation and being able to compare impact over time.
- **Limitations:** Per capita values may obscure the size of emissions in countries with large populations.

I cleaned and transformed the data so that it could be visualized properly.

Country Name	Country Code	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001
Aruba	ABW	3.432681802	3.631620897	3.411820588	3.680649526	3.937096673	2.488526999	3.970985272	3.967168263	3.965023032	2.965072637	2.965072637
Africa Eastern and Southern	AFE	1.174578007	1.123987955	1.093278599	1.088835819	1.11724841	1.128709727	1.145021672	1.136120713	1.073663127	1.062240409	1.062240409
Afghanistan	AFG	0.232619344	0.137051251	0.119439166	0.103699548	0.085726829	0.079782626	0.075037566	0.074133627	0.069163057	0.05047608	0.05047608
Africa Western and Central	AFW	0.501841936	0.528290038	0.506310369	0.477636486	0.507183274	0.571195525	0.553488161	0.518728097	0.511601787	0.533661233	0.533661233
Angola	AGO	0.979787216	0.975473353	0.946555875	0.867215446	0.945511672	1.106402503	1.118802042	1.123819789	1.150898675	1.019576015	1.019576015
Albania	ALB	1.34061265	0.72773995	0.688720898	0.730810192	0.649353909	0.64863592	0.502019991	0.61201267	0.992866007	1.046446017	1.046446017
Arab World	ARB	3.072479722	3.192236679	3.321922722	3.433509575	3.448098606	3.453320596	3.531726351	3.491071953	3.464533454	3.506669346	3.506669346
United Arab Emirates	ARE	32.2542417	29.56519272	29.66306991	31.16951896	31.79188733	31.1410229	29.92166564	28.44280447	27.29704664	25.31988007	25.31988007
Argentina	ARG	3.177606554	3.190850278	3.258139023	3.313254361	3.315321748	3.563984281	3.642126438	3.794254837	3.803231674	3.675005611	3.675005611
Armenia	ARM	5.976269356	3.073075912	1.481265051	0.8346584	1.070631377	0.791189285	1.028374245	1.082346004	0.980082001	1.108192853	1.108192853
American Samoa	ASM	0.002088467	0.002039235	0.00199366	0.001951181	0.001911461	0.001874309	0.00183952	0.001807207	0.001777841	0.00175886	0.00175886
Antigua and Barbuda	ATG	2.593591746	2.569716167	2.45932883	2.628547492	2.696915925	2.62023789	2.716282212	2.728829476	2.717671653	1.807454079	1.807454079
Australia	AUS	16.18450691	16.19765502	16.27598667	16.42055126	16.81167919	17.15639492	17.49824961	18.38544434	18.50456171	18.59653067	18.59653067
Austria	AUT	8.446334062	7.717350561	7.660967819	7.697882012	8.058739264	8.540300894	8.475144643	8.53620423	8.286125537	8.324302639	8.324302639
Azerbaijan	AZE	7.541168704	6.082497409	5.254963441	4.543114967	4.408661197	3.68611362	3.49825535	3.459737141	3.431474116	3.577566285	3.577566285
Burundi	BDI	0.049598018	0.038030816	0.043668638	0.044943395	0.041672738	0.044757389	0.044760811	0.046354452	0.046598197	0.045547321	0.045547321
Belgium	BEL	11.96775127	11.70985066	11.47275391	12.07156904	12.07643114	12.544418	12.22036205	12.48692542	12.03125943	12.19066943	12.19066943
Benin	BEN	0.064054407	0.067517005	0.065404262	0.064020277	0.063708762	0.168073196	0.207158497	0.193593362	0.205674344	0.213927099	0.213927099
Burkina Faso	BFA	0.048878977	0.040813041	0.041755671	0.044360822	0.044412472	0.047982619	0.074728675	0.068608047	0.068898917	0.073690546	0.073690546
Bangladesh	BGD	0.123632634	0.135672295	0.143494128	0.149566433	0.177625891	0.17686861	0.188333834	0.188840145	0.19778202	0.20048266	0.20048266

Figure 2: Cleaned and transformed CO2 data.

1. CO2 Emissions by Country Name and Year (Stacked Bar Chart)

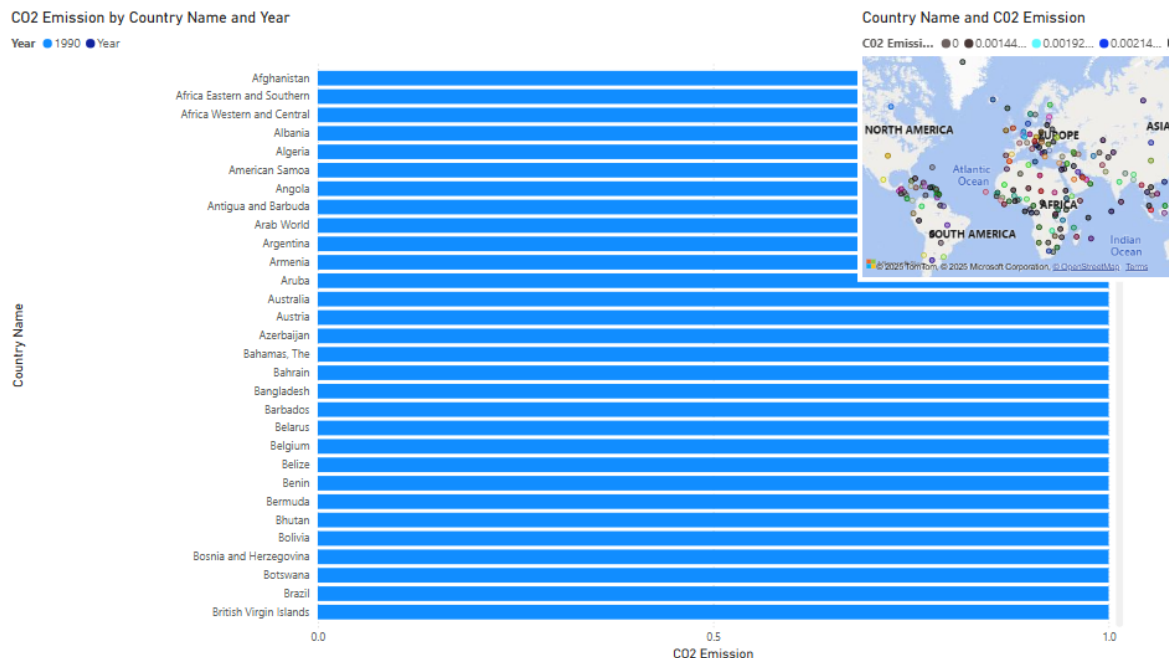


Figure 3: Stacked Bar Chart of CO2 Emissions by Country Name and Year.

- This dataset shows CO2 emissions ranked by country.
- CO2 Emissions: quantify CO2 released due to activities such as burning fossil fuels or combustible materials for transportation or industry, as indicated by CO2 emissions.
- Metric tons per capita: a standardized unit of per-person CO2 emissions for a given country.
- Country names are included on the Y-axis, such as Afghanistan, Albania, Algeria, Angola, Argentina, Australia, Austria, Brazil, Belgium, and other countries.
- The X-axis shows the emission values.

The Bar Chart can be used to compare CO2 emissions across a handful of countries and allows us to identify regions of the world with high and low emissions. The Map Visualization provides a way to visualize the emission intensity among the countries, which can help to frame global differences in emissions.

Limitations in the Data Set

- Restricted Time Range: The graph represents emissions from 1990, without enough data for recent years or long-term trends.
- Missing clear unit definition: The quantities of CO2 emissions measurement are unclear.

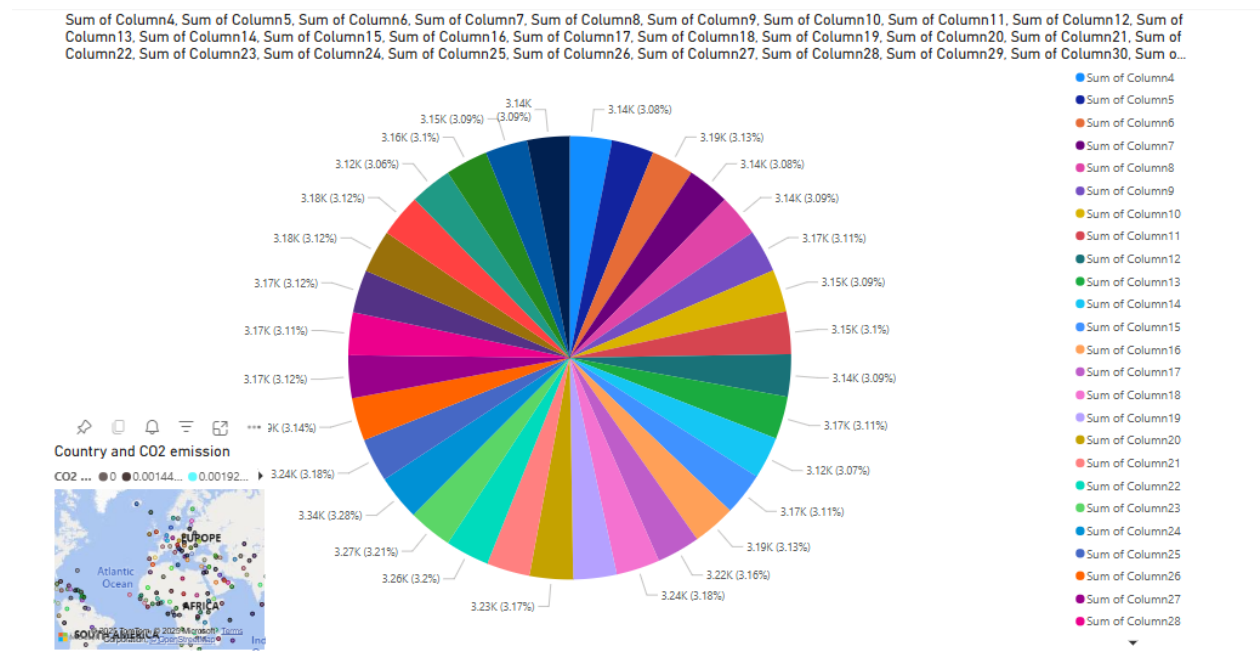


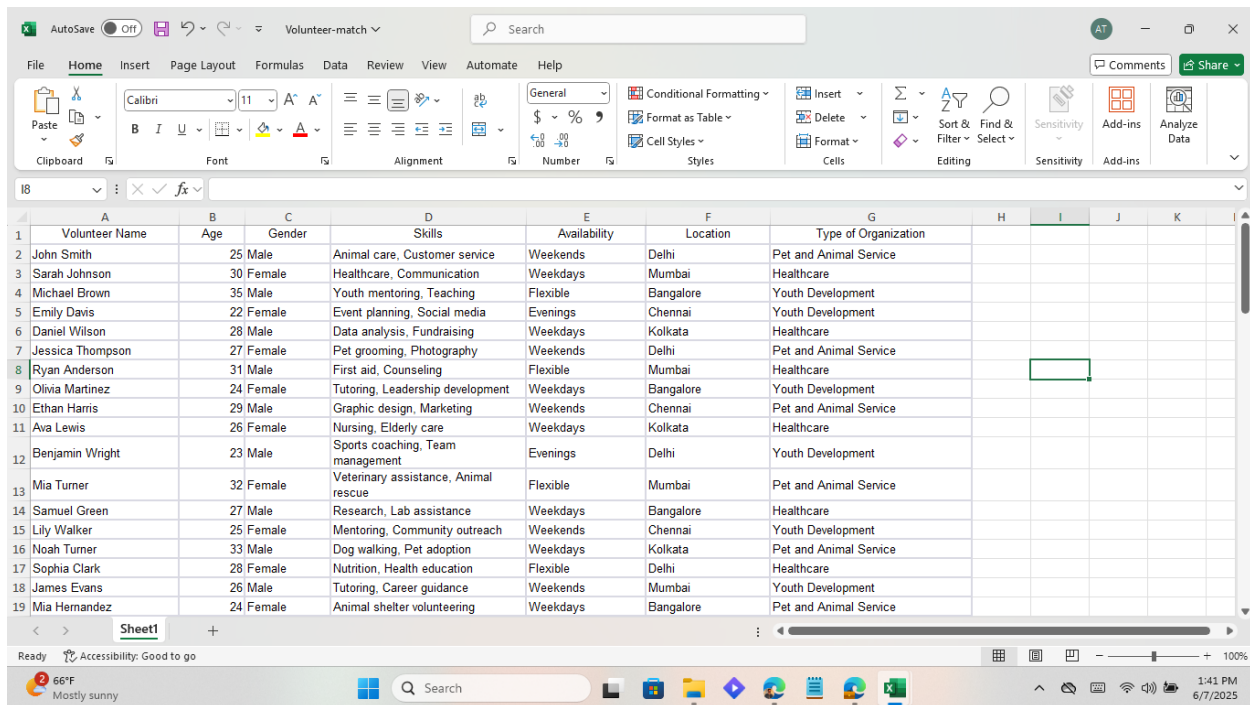
Figure 4: Pie Chart of CO2 Emissions by year.

- A Pie Chart that enables comparison of multiple column sums, making it easy to see their relative distribution.
- The map provides a spatial representation of CO2 emissions to characterize the geography of emissions.
- All the years are separated in the column - the CO2 emission value is in percentage terms.
- The pie chart is the same color as the corresponding column.
- The markers on the map provide the global location of countries.
- CO2 Emission Levels markers show CO2 levels.
- CO2 emissions reported in metric tons.
- Different marker colors reflect the different emission levels.
- Carbon dioxide emissions from industry, energy, and transportation.
- Percentage Distribution of the column values in proportion terms.

Limitations in the Data Set

- While regions are mapped, total global emissions are not compared.

Dataset 2: Global Volunteer Engagement



	A	B	C	D	E	F	G	H	I	J	K
	Volunteer Name	Age	Gender	Skills	Availability	Location	Type of Organization				
1	John Smith	25	Male	Animal care, Customer service	Weekends	Delhi	Pet and Animal Service				
2	Sarah Johnson	30	Female	Healthcare, Communication	Weekdays	Mumbai	Healthcare				
3	Michael Brown	35	Male	Youth mentoring, Teaching	Flexible	Bangalore	Youth Development				
4	Emily Davis	22	Female	Event planning, Social media	Evenings	Chennai	Youth Development				
5	Daniel Wilson	28	Male	Data analysis, Fundraising	Weekdays	Kolkata	Healthcare				
6	Jessica Thompson	27	Female	Pet grooming, Photography	Weekends	Delhi	Pet and Animal Service				
7	Ryan Anderson	31	Male	First aid, Counseling	Flexible	Mumbai	Healthcare				
8	Olivia Martinez	24	Female	Tutoring, Leadership development	Weekdays	Bangalore	Youth Development				
9	Ethan Harris	29	Male	Graphic design, Marketing	Weekends	Chennai	Pet and Animal Service				
10	Ava Lewis	26	Female	Nursing, Elderly care	Weekdays	Kolkata	Healthcare				
11	Benjamin Wright	23	Male	Sports coaching, Team management	Evenings	Delhi	Youth Development				
12	Mia Turner	32	Female	Veterinary assistance, Animal rescue	Flexible	Mumbai	Pet and Animal Service				
13	Samuel Green	27	Male	Research, Lab assistance	Weekdays	Bangalore	Healthcare				
14	Lily Walker	25	Female	Mentoring, Community outreach	Weekends	Chennai	Youth Development				
15	Noah Turner	33	Male	Dog walking, Pet adoption	Weekdays	Kolkata	Pet and Animal Service				
16	Sophia Clark	28	Female	Nutrition, Health education	Flexible	Delhi	Healthcare				
17	James Evans	26	Male	Tutoring, Career guidance	Weekends	Mumbai	Youth Development				
18	Mia Hernandez	24	Female	Animal shelter volunteering	Weekdays	Bangalore	Pet and Animal Service				

Figure 5: Dataset of Global Volunteer Engagement (Yadav, 2023).

a. Who collected the data?

- Collected via **Gallup World Poll**, data hosted on **Kaggle**.
- Link: [Volunteer NGO Match Dataset](#)

b. Variables included:

- Country
- Percent of population volunteering

- Year
- Volunteer types (formal/informal)
- Gender breakdown

c. Size of dataset:

- 5,000–10,000 records across 100+ countries
- Variables: Around 6-8 columns

d. Locations included:

- Global dataset: regions from Asia, Africa, the Americas, Europe

e. Relevance and Limitations:

- **Supports:** Uses real data from a completed survey-based, survey-linked econometric model measuring volunteer engagement, which is built on real survey evidence and educated assumptions and falls within the author's area of expertise.
- **Limitations:** Survey-based; some countries have missing or stale entries.

1. Age by Location and Type of Organization (Ribbon Chart)

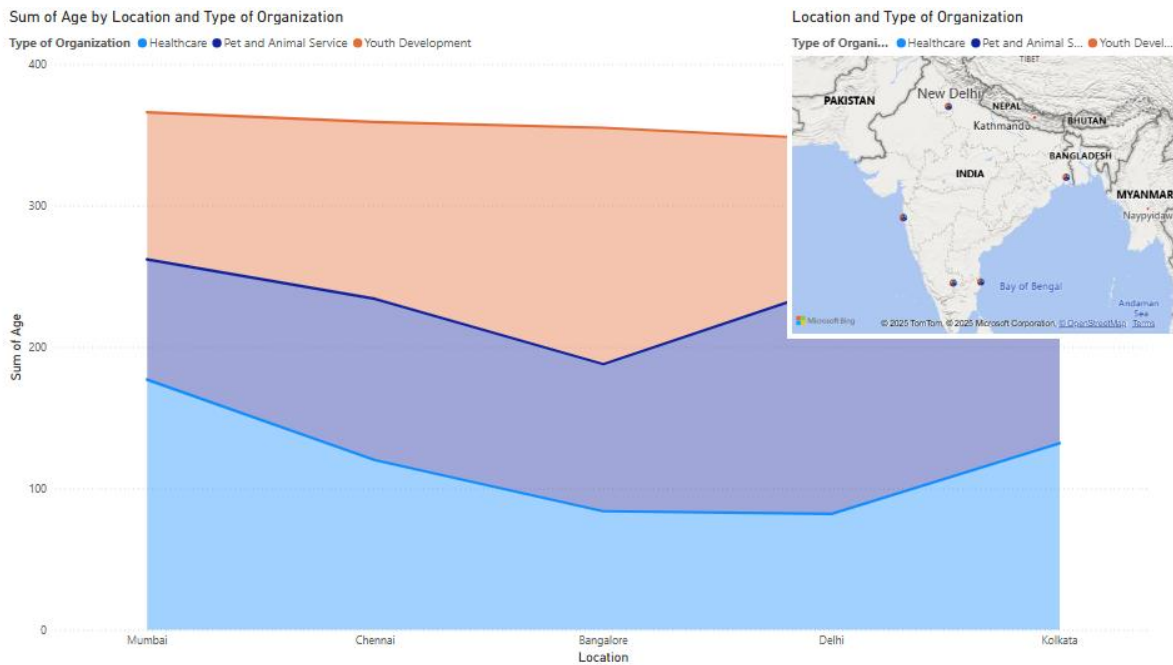


Figure 6: Ribbon Chart of Age by Location and Type of Organization.

- Shows the total age represented by age across categories of organization in the five largest cities in India.
- Depicts how categories of organization vary by city in terms of total age.
- Provides a representation of the geographic spread of the several types of organizations.
- Depicts the total age of individuals in each type of organization.
- The type of Organization is classified as Healthcare, Pet and Animal Service, and Youth Development.

Limitations in the Data Set

- No context on sample size, unclear how many individuals contribute to age sums.

The Line Chart illustrates trends in summed ages for organization types for different cities, and the Map shows where these organizations are concentrated geographically.

2. Availability by Volunteer Name (Stacked Bar Chart)

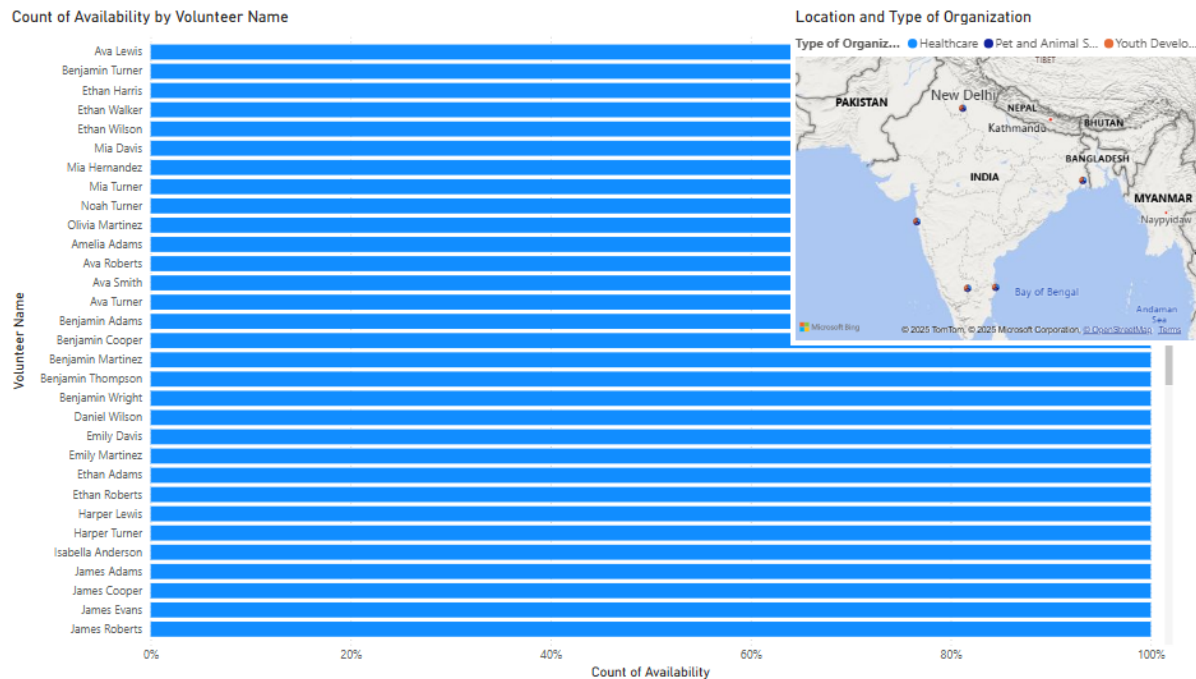


Figure 7: Stacked Bar Chart of Availability by Volunteer Name.

- Tracks how many volunteers are available as a percentage.
- The chart shows that all volunteers indicated 100% of their time available.
- Organization types are Categories and represented by colors:
- The Map visually shows the several types of organizations grouped geographically.
- Availability maps how much of the time volunteers are free for assignments.
- Types of organizations are categorized by the services they offer (For example, Healthcare, Animal Services, and Youth Development).

Limitations in the Data Set

- The Data coverage will not be complete. It could miss smaller organizations or new volunteers due to registration in only a one-month period.

The Volunteer Availability Chart below makes it quite simple to determine if you are looking for volunteers in a specific period who have full availability. The map of Organizations below also helps visually see where the several types of organizations are grouped, which could help when deciding who to assign volunteers.

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